



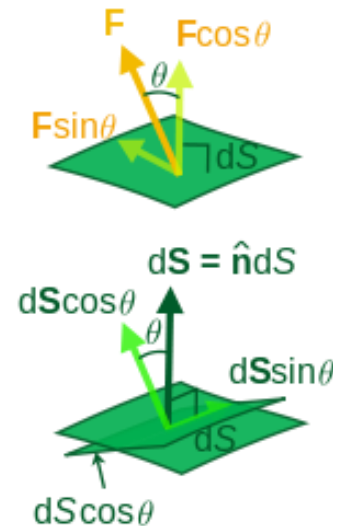
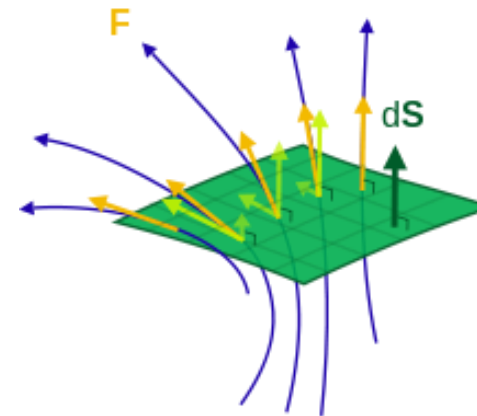
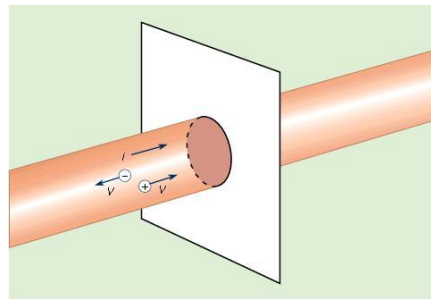
Engineering Electromagnetics

工程电磁场

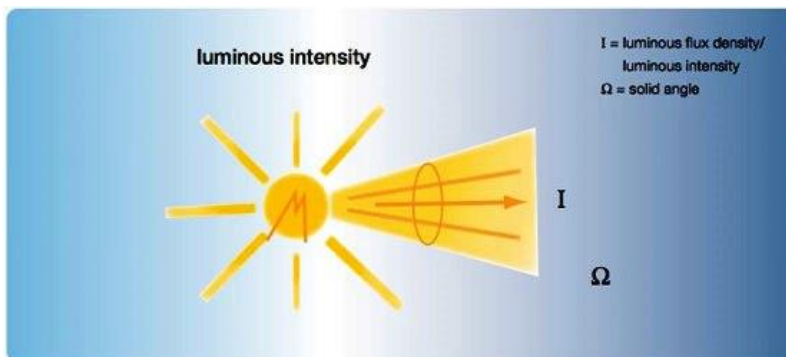
Youwei Jia 嘉有为

4: Concept of 'Flux' 通量

- The word 'flux' comes from Latin: fluxus means "flow", and fluere is "to flow". 一个表征“流动”的量
- Flux is defined as the rate of flow of a **property** through certain area. “通量”用来表示单位时间内流过特定面的**某个量**的多少



$$\begin{aligned} \mathbf{F} \cdot d\mathbf{S} &= (F \cos \theta) dS \\ &= F (dS \cos \theta) \end{aligned}$$



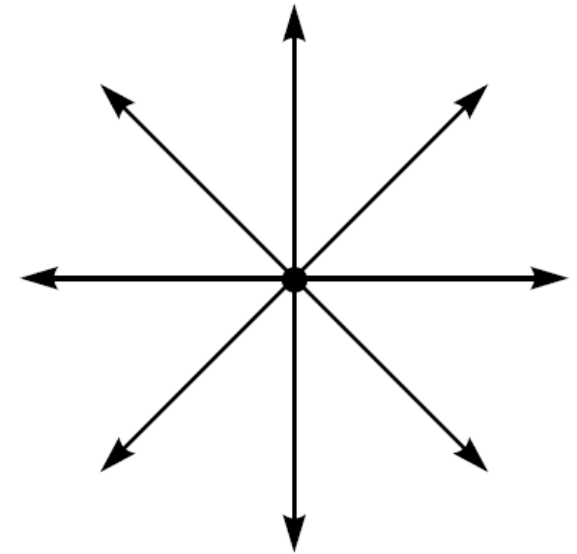
5: Electric Flux Density 电通密度



- Streamlines, Electric flux lines 电力线、电通量线

Let us be content to show only the *direction* of $\mathbf{E}$ by drawing **continuous lines**, which are everywhere **tangent to $\mathbf{E}$** , from the charge.

These lines are usually called **streamlines**, although other terms such as **flux lines** 通量线 and **direction lines** 方向线 are also used.



Q6. Is it really that the magnitude of field can not be indicated with streamlines?

Q1. How can we define 'flux' when there is nothing projected outward from the point charge?

Note: There should be 'something' to indicate the 'influence' of this point charge on the space surrounding it. That is why we define 'electric flux density'

5: Electric Flux Density 电通密度



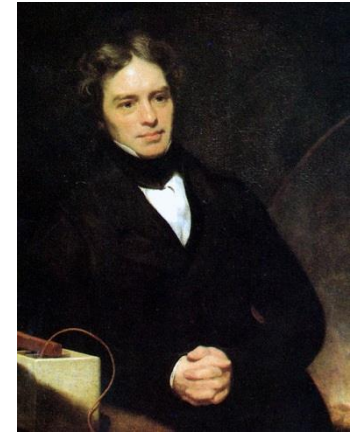
- Faraday Experiment

He started with a pair of metal spheres of different sizes; the larger one consisted of two hemispheres that could be assembled around the smaller sphere.

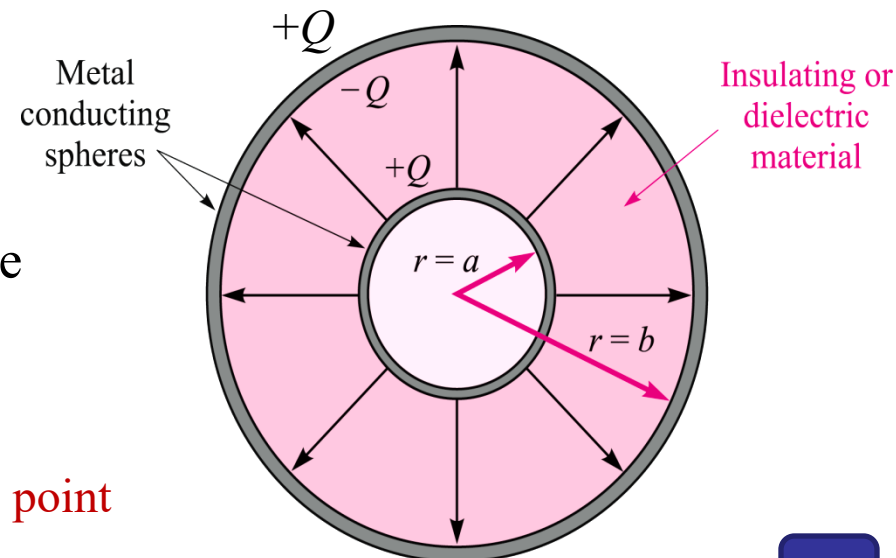
The inner charge, Q , **induces** an equal and opposite charge, $-Q$, on the inside surface of the outer sphere, by attracting free electrons in the outer material toward the positive charge.

This means that before the outer sphere is grounded, charge $+Q$ resides on the *outside* surface of the outer conductor.

Note: Herein, ‘induce’ is the ‘influence’ of this point charge on the outer metal sphere.



Michael Faraday, (1791 – 1867)

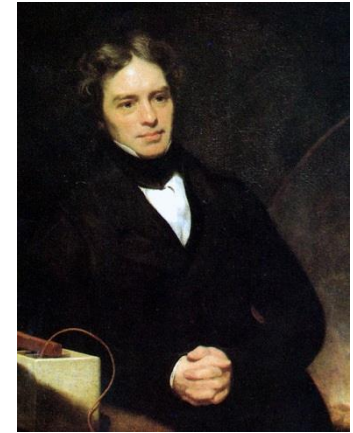


5: Electric Flux Density 电通密度



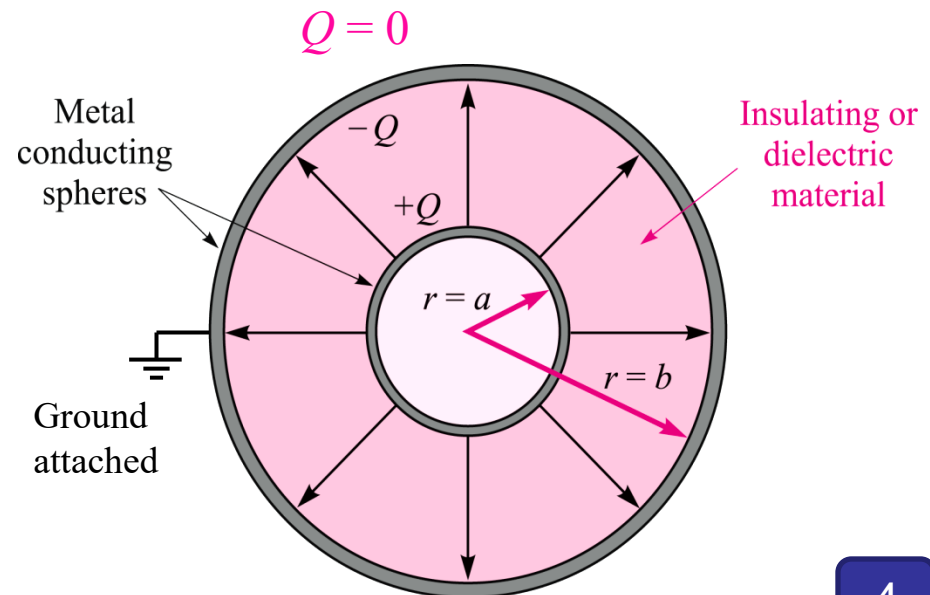
- Faraday Experiment

Attaching the ground connects the outer surface to an unlimited supply of free electrons, which then **neutralize 中和** the positive charge layer. The net charge on the outer sphere is then the charge on the inner layer, or $-Q$.



Michael Faraday, (1791 – 1867)

Faraday found that the total charge on the outer sphere was equal in *magnitude* to the original charge placed on the inner sphere and that this was true regardless of the dielectric material separating the two spheres and the radius of the outer metal sphere.



5: Electric Flux Density 电通密度



- Faraday Experiment- Electric Flux

Faraday concluded that there occurred a charge “**displacement**” from the inner sphere to the outer sphere.

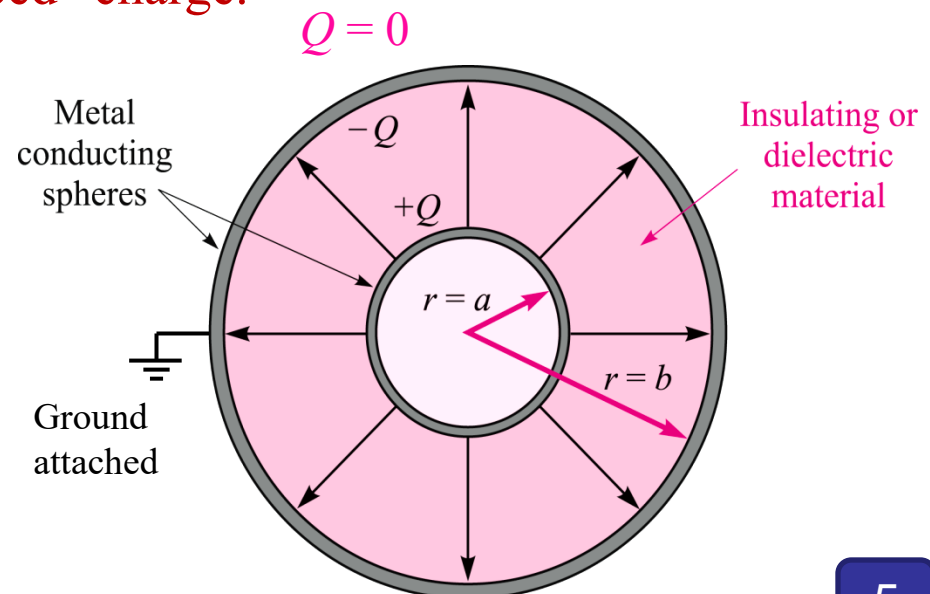
Displacement involves a *flow* or *flux*, Ψ , existing within the dielectric, and whose magnitude is equivalent to the amount of “displaced” charge.

Specifically:

$$\Psi = Q$$



Michael Faraday, (1791 – 1867)



5: Electric Flux Density 电通密度



- Faraday Experiment-Electric Flux Density

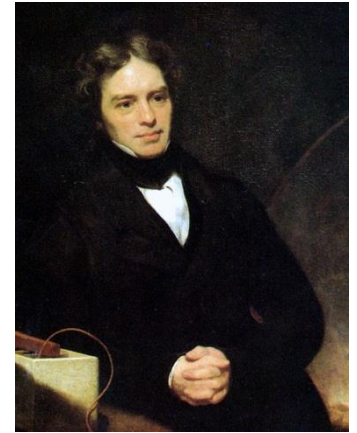
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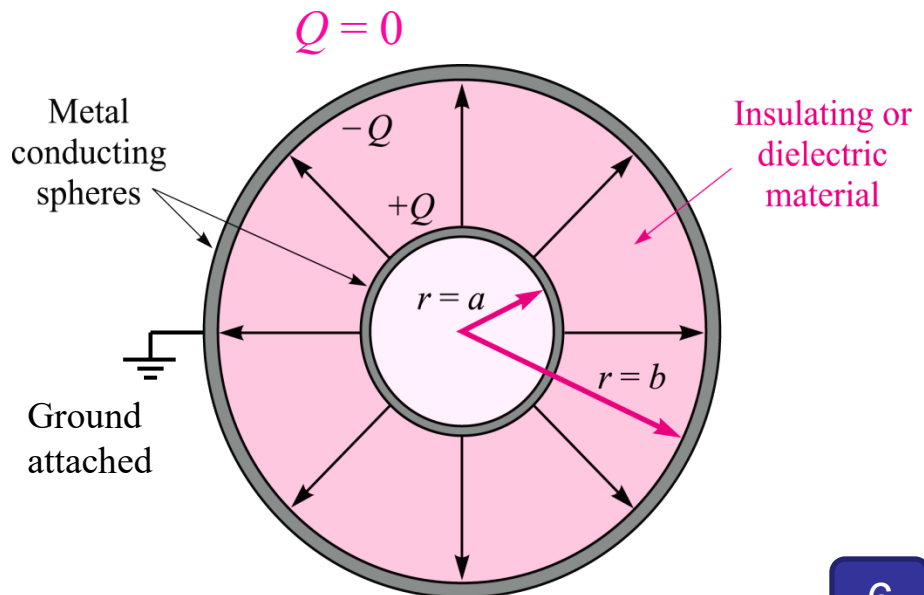
$$\Psi = Q$$

The **density of flux** at the inner sphere surface is equivalent to the density of charge there (in Coul/m²)

$$D(r = a) = \frac{\Psi}{4\pi a^2} = \frac{Q}{4\pi a^2}$$



Michael Faraday, (1791 – 1867)



5: Electric Flux Density 电通密度

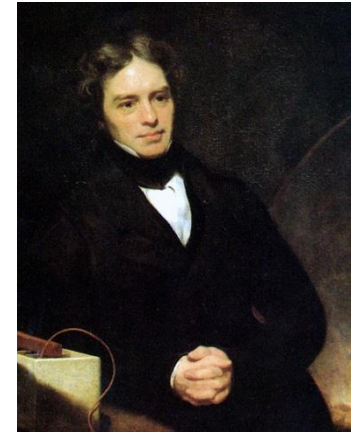


- Vector Field Description of Flux Density

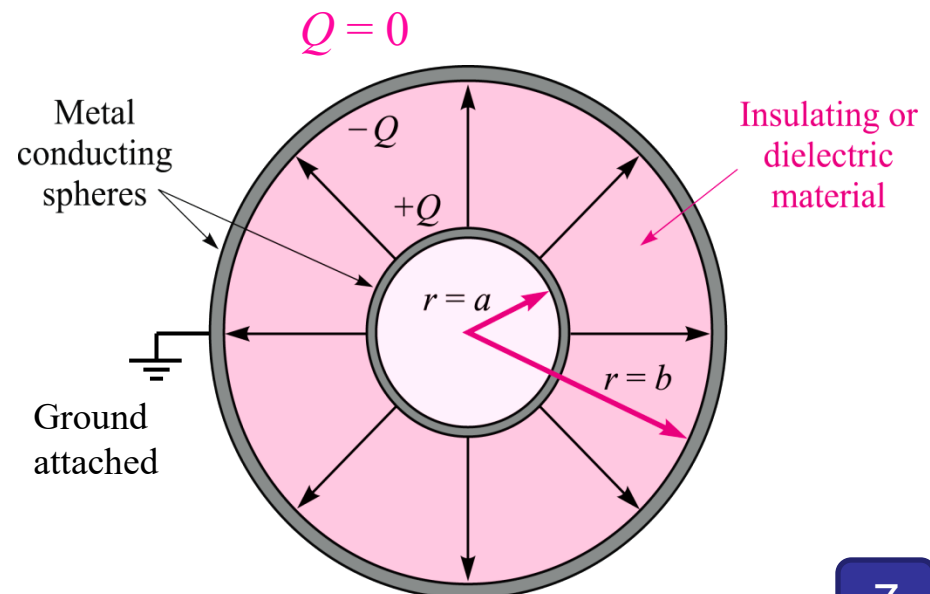
A vector field is established which points in the direction of the “flow” or displacement. In this case, the direction is the outward radial direction in spherical coordinates. At each surface, we would have:

$$\mathbf{D} \Big|_{r=a} = \frac{Q}{4\pi a^2} \mathbf{a}_r \quad (\text{inner sphere})$$

$$\mathbf{D} \Big|_{r=b} = \frac{Q}{4\pi b^2} \mathbf{a}_r \quad (\text{outer sphere})$$



Michael Faraday, (1791 – 1867)



5: Electric Flux Density 电通密度

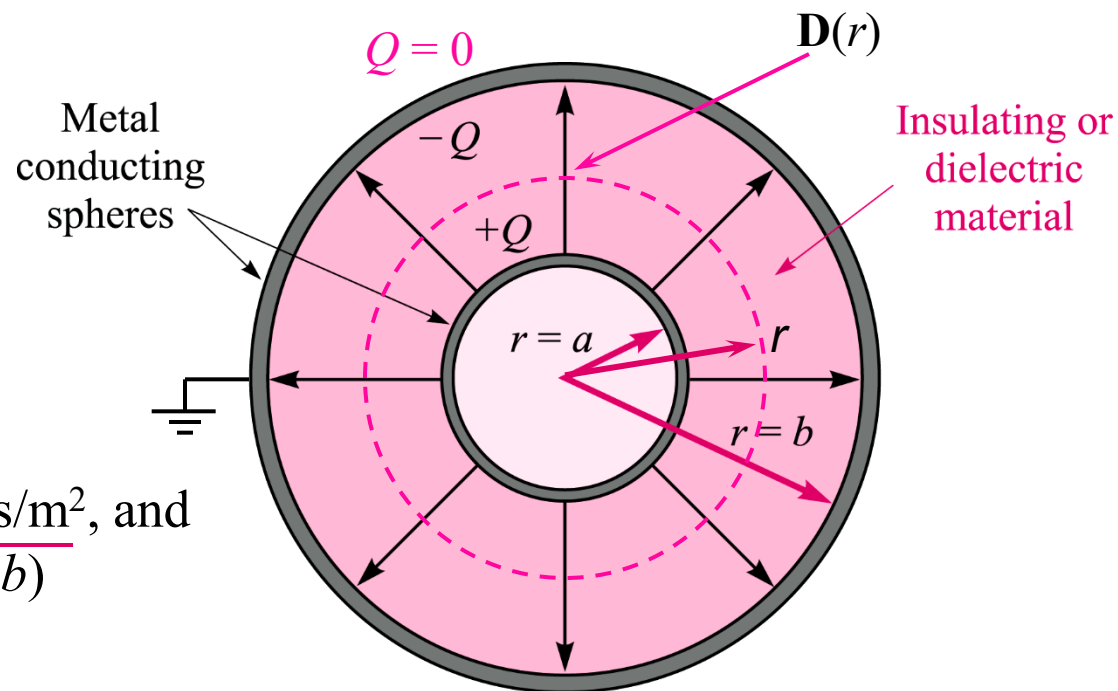


- Vector Field Description of Flux Density

At a general radius r between spheres, we would have:

$$\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$$

Expressed in units of Coulombs/m², and defined over the range $(a \leq r \leq b)$



5: Electric Flux Density 电通密度



- Point Charge Fields – Relationship between **E** and **D**

If we now let the inner sphere radius reduce to a point, while maintaining the same charge, and let the outer sphere radius approach infinity, we have a point charge. The electric flux density is unchanged, but is defined over all space:

$$\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r \quad \text{C/m}^2 \quad (0 < r < \infty)$$

We compare this to the electric field intensity in free space:

$$\mathbf{E} = \frac{Q}{4\pi \epsilon_0 r^2} \mathbf{a}_r \quad \text{V/m} \quad (0 < r < \infty)$$

..and we see that:

$$\mathbf{D} = \epsilon_0 \mathbf{E} \quad \text{(free space only)}$$

5: Electric Flux Density 电通密度



- Volume Charge Fields – Relationship between **E** and **D**

We learned in Chapter 2 that:

$$\mathbf{E} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad \text{(free space only)}$$

It now follows that:

$$\mathbf{D} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi R^2} \mathbf{a}_R$$

..and we also see that:

$$\mathbf{D} = \epsilon_0 \mathbf{E} \quad \text{(free space only)}$$

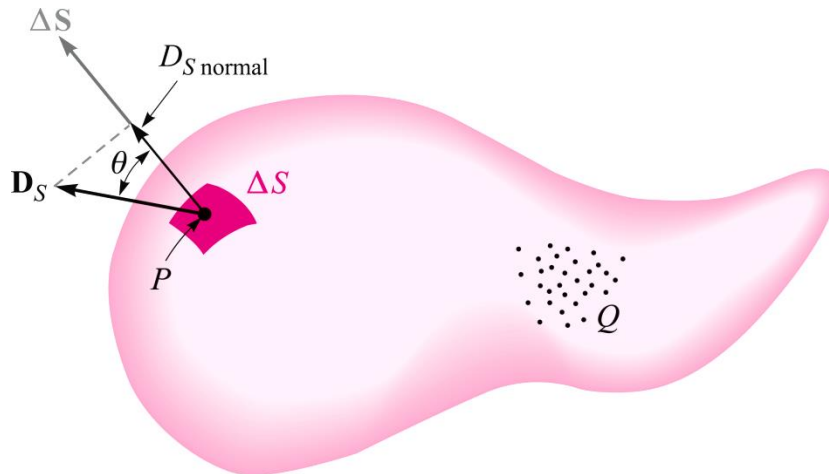
6: Gauss' Law 高斯定律



- Gauss' Law

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface

穿过任意闭合面的电通量等于该曲面所包含的总电荷



Carl Friedrich Gauss (1777 -1855)

6: Gauss' Law 高斯定律



- Gauss' Law- Its mathematical statement

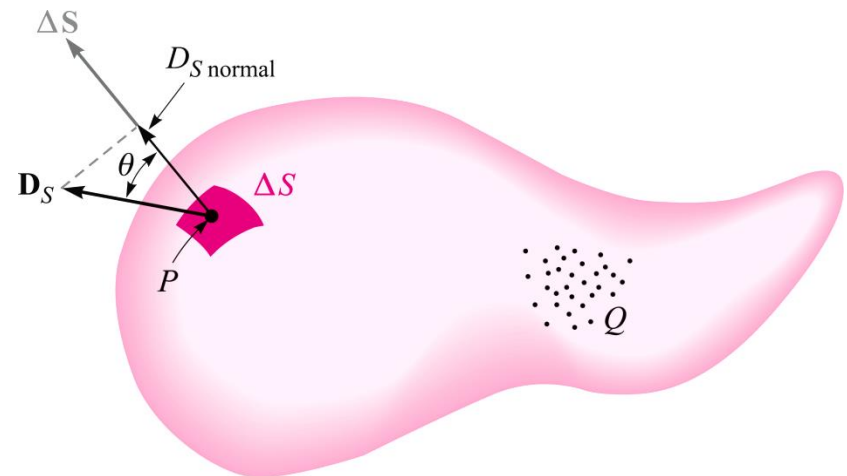
The electric flux passing through any closed surface is equal to the total charge enclosed by that surface
穿过任意闭合面的电通量等于该曲面所包含的总电荷

$$\Delta \Psi = \text{flux crossing } \Delta S = D_{S,\text{norm}} \Delta S = D_S \cos \theta \Delta S = \mathbf{D}_S \cdot \Delta \mathbf{S}$$

We define the **differential surface area** (a vector) as

$$d\mathbf{S} = \mathbf{n} dS$$

where **$\mathbf{n}$** is the **unit outward normal vector to the surface**, and where dS is the area of the differential spot on the surface



$$\Psi = \int d\Psi = \oint_{\text{closed surface}} \mathbf{D}_S \cdot d\mathbf{S}$$

6: Gauss' Law 高斯定律



- Gauss' Law- Its mathematical statement

$$\Psi = \oint_S \mathbf{D}_S \cdot d\mathbf{S} = \text{charge enclosed} = Q$$

in which the charge can exist in the form of point charges: $Q = \sum Q_m$

or a continuous charge distribution:

Line charge: $Q = \int \rho_L dL$

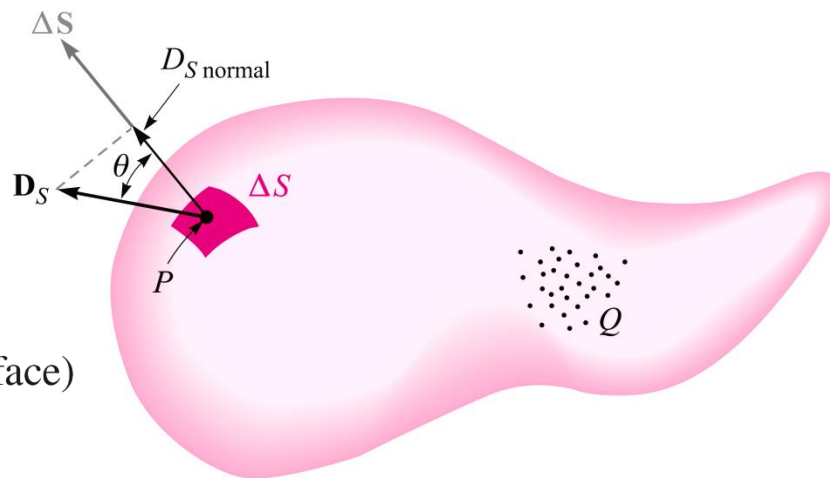
Surface charge:

$$Q = \int_S \rho_S dS \quad (\text{not necessarily a closed surface})$$

Volume charge: $Q = \int_{\text{vol}} \rho_v dv$

For a volume charge, we would have:

$$\oint_S \mathbf{D}_S \cdot d\mathbf{S} = \int_{\text{vol}} \rho_v dv$$

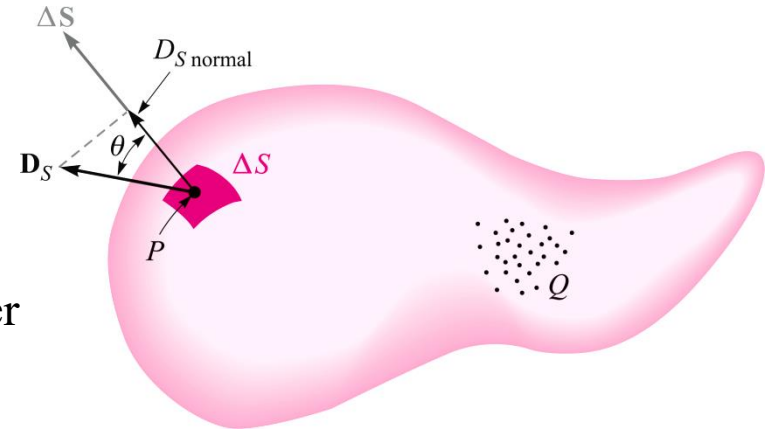


6: Gauss' Law 高斯定律



- Using Gauss' Law to solve for **D** evaluated at a surface

$$Q = \oint_S \mathbf{D}_S \cdot d\mathbf{S} \quad \text{where } d\mathbf{S} = \mathbf{n} dS$$



The solution is easy if we can choose a surface, S , over which to integrate (Gaussian surface) that satisfies the following two conditions:

- $\mathbf{D}_S$ is everywhere either normal or tangential to the closed surface, so that $\mathbf{D}_S \cdot d\mathbf{S}$ becomes either $D_S dS$ or zero, respectively.
- On that portion of the closed surface for which $\mathbf{D}_S \cdot d\mathbf{S}$ is not zero, $D_S = \text{constant}$.

The integral now simplifies:

$$\oint_S \mathbf{D}_s \cdot d\mathbf{S} = \underbrace{\oint_S D_s dS}_{\text{Condition 1}} = \underbrace{D_s \oint_S dS}_{\text{Condition 2}} = Q$$

So that:

$$D_s = \frac{Q}{\oint_S dS}$$

6: Gauss' Law 高斯定律



- Example: Point Charge Field

Begin with the radial flux density: $\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$

and consider a spherical surface of radius a that surrounds the charge, on which:

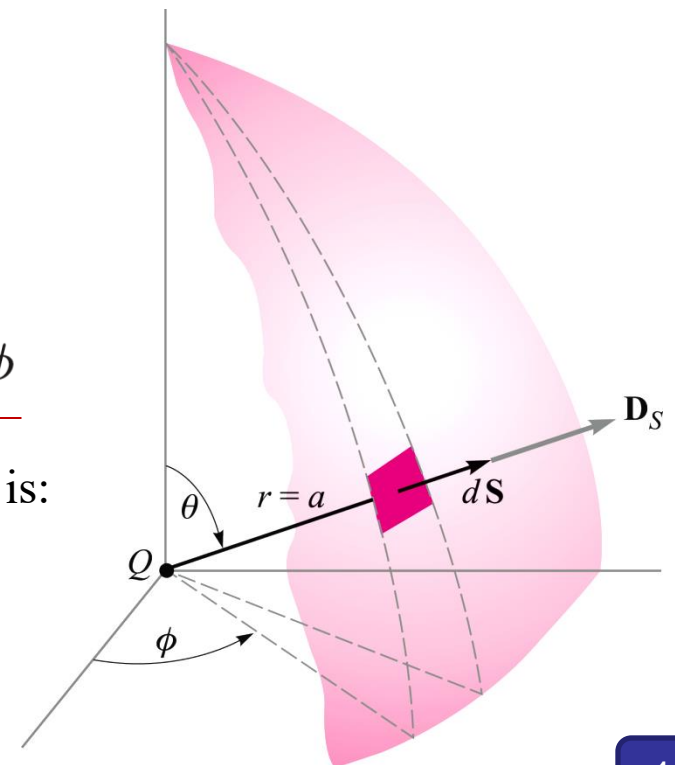
$$\mathbf{D}_S = \frac{Q}{4\pi a^2} \mathbf{a}_r$$

On the surface, the differential area is:

$$dS = r^2 \sin \theta d\theta d\phi = a^2 \sin \theta d\theta d\phi$$

and this, combined with the outward unit normal vector is:

$$d\mathbf{S} = a^2 \sin \theta d\theta d\phi \mathbf{a}_r$$



6: Gauss' Law 高斯定律



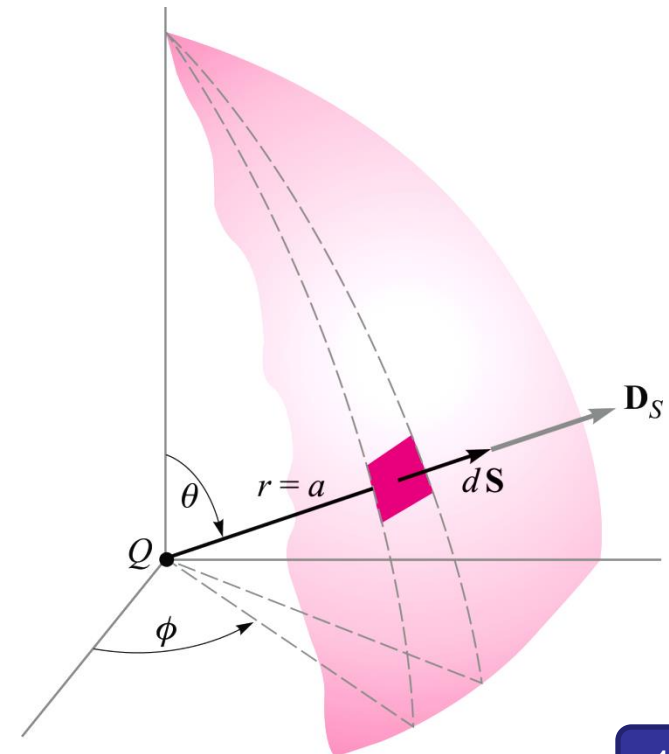
- Example: Point Charge Field

Now, the integrand becomes:

$$\mathbf{D}_S \cdot d\mathbf{S} = \frac{Q}{4\pi a^2} a^2 \sin \theta d\theta d\phi \mathbf{a}_r \cdot \mathbf{a}_r = \frac{Q}{4\pi} \sin \theta d\theta d\phi$$

and the integral is set up as:

$$\begin{aligned} \oint_S \mathbf{D}_S \cdot d\mathbf{S} &= \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} \frac{Q}{4\pi} \sin \theta d\theta d\phi \\ &= \int_0^{2\pi} \frac{Q}{4\pi} (-\cos \theta)_0^\pi d\phi = \int_0^{2\pi} \frac{Q}{2\pi} d\phi = \underline{Q} \end{aligned}$$



6: Gauss' Law 高斯定律



- Example: Line Charge Field

Consider a line charge of uniform charge density ρ_L on the z axis that extends over the range $-\infty < z < \infty$.

We need to choose **an appropriate Gaussian surface**, being mindful of these considerations:

1. With which coordinates does the field vary (or of what variables is D a function)?
2. Which components of $\mathbf{D}$ are present?

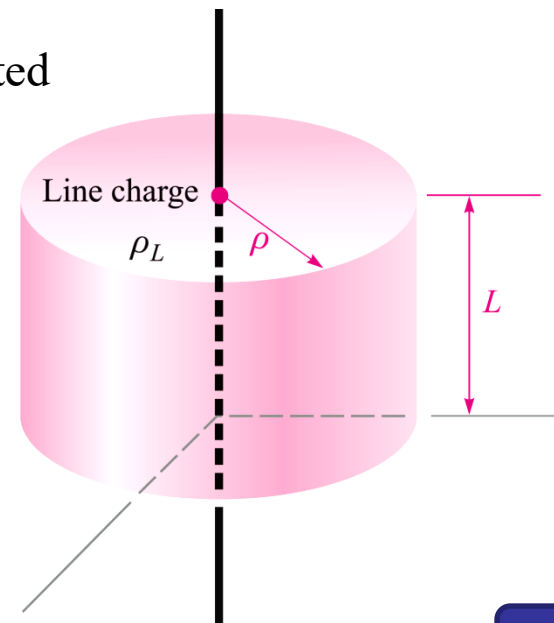
We know from symmetry that the field will be radially-directed (normal to the z axis) in cylindrical coordinates:

$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$

and that the field will vary with radius only:

$$D_\rho = f(\rho)$$

So we choose a cylindrical surface of radius ρ , and of length L .



6: Gauss' Law 高斯定律

- Example: Line Charge Field

We apply Gauss's law,

$$\begin{aligned} Q &= \oint_{\text{cyl}} \mathbf{D}_S \cdot d\mathbf{S} = D_S \int_{\text{sides}} dS + 0 \int_{\text{top}} dS + 0 \int_{\text{bottom}} dS \\ &= D_S \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho d\phi dz = D_S 2\pi \rho L \end{aligned}$$

and obtain

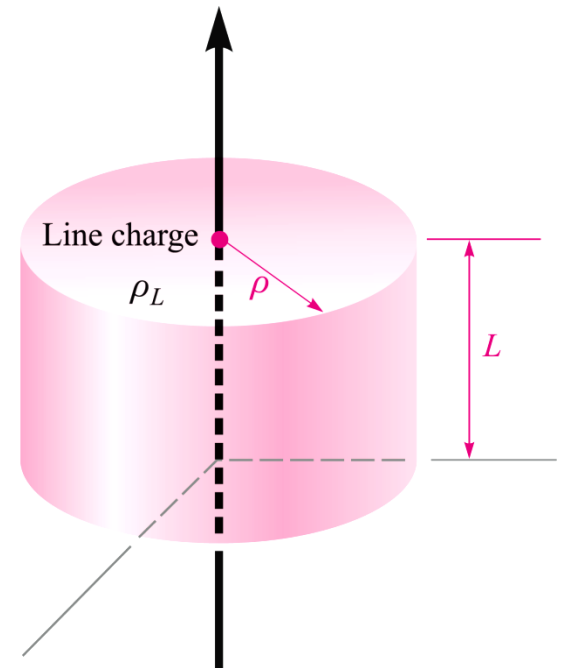
$$D_S = D_\rho = \frac{Q}{2\pi \rho L}$$

In terms of the charge density ρ_L , the total charge enclosed is

$$Q = \rho_L L$$

Giving: $D_\rho = \frac{\rho_L}{2\pi \rho}$

So that finally: $E_\rho = \frac{\rho_L}{2\pi \epsilon_0 \rho}$



Note: Please compare the two methods of solving line charge field intensity E , one by using Gauss's law and the other by using Coulomb's law.

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



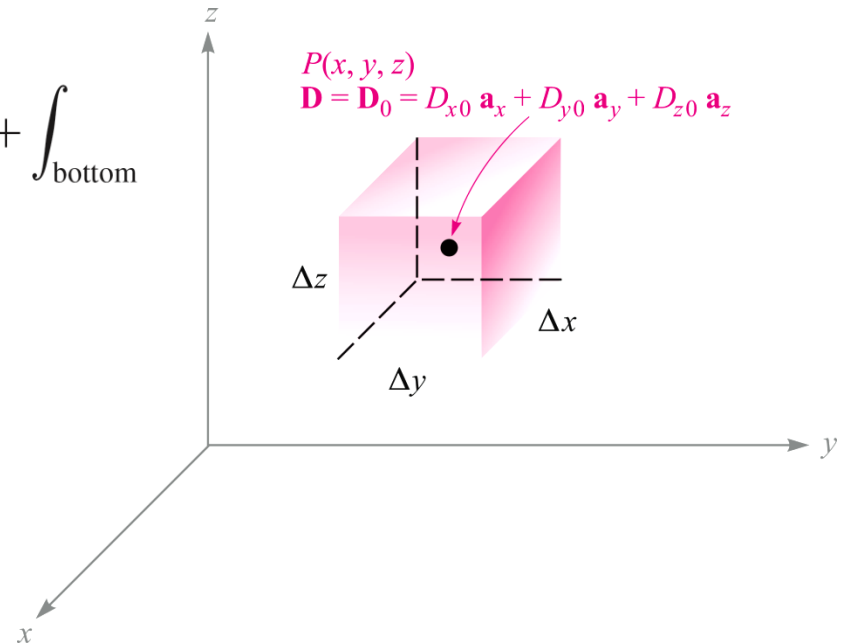
- Electric Flux Within a Differential Volume Element

The value of $\mathbf{D}$ at the point P may be expressed in rectangular components, $\mathbf{D}_0 = D_{x0}\mathbf{a}_x + D_{y0}\mathbf{a}_y + D_{z0}\mathbf{a}_z$. We choose as our closed surface the small rectangular box, centered at P , having sides of lengths Δx , Δy , and Δz , and apply Gauss's law,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{front}} + \int_{\text{back}} + \int_{\text{left}} + \int_{\text{right}} + \int_{\text{top}} + \int_{\text{bottom}}$$

Taking the front surface, for example, we have:

$$\begin{aligned} \int_{\text{front}} &\doteq \mathbf{D}_{\text{front}} \cdot \Delta \mathbf{S}_{\text{front}} \\ &\doteq \mathbf{D}_{\text{front}} \cdot \Delta y \Delta z \mathbf{a}_x \\ &\doteq D_{x,\text{front}} \Delta y \Delta z \end{aligned}$$



7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



• Electric Flux Within a Differential Volume Element

Taking the front surface, for example, we have:

$$\begin{aligned}\int_{\text{front}} &\doteq \mathbf{D}_{\text{front}} \cdot \Delta \mathbf{S}_{\text{front}} \\ &\doteq \mathbf{D}_{\text{front}} \cdot \Delta y \Delta z \mathbf{a}_x \\ &\doteq D_{x,\text{front}} \Delta y \Delta z\end{aligned}$$

$$\begin{aligned}D_{x,\text{front}} &\doteq D_{x0} + \frac{\Delta x}{2} \times \text{rate of change of } D_x \text{ with } x \\ &\doteq D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}\end{aligned}$$

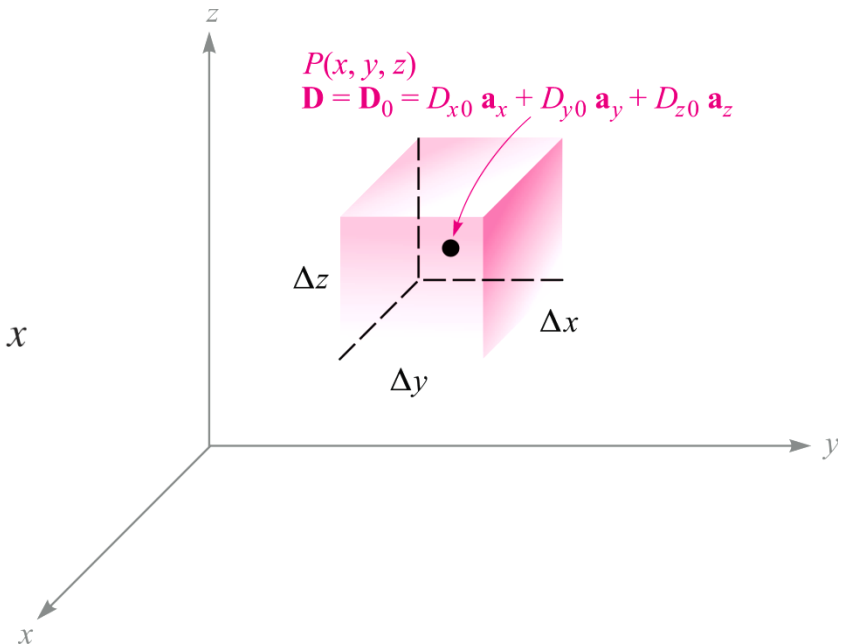
We now have: $\int_{\text{front}} \doteq \left(D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$

and in a similar manner:

$$\int_{\text{back}} \doteq \left(-D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

Therefore:

$$\int_{\text{front}} + \int_{\text{back}} \doteq \frac{\partial D_x}{\partial x} \Delta x \Delta y \Delta z$$



— minus sign because D_{x0} is *inward* flux through the back surface.

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程

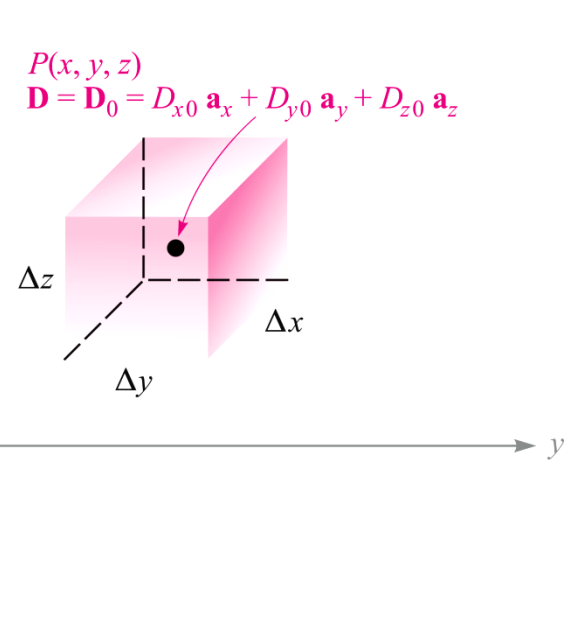


- Electric Flux Within a Differential Volume Element

Now, by a similar process, we find that:

$$\int_{\text{right}} + \int_{\text{left}} \doteq \frac{\partial D_y}{\partial y} \Delta x \Delta y \Delta z$$

and $\int_{\text{top}} + \int_{\text{bottom}} \doteq \frac{\partial D_z}{\partial z} \Delta x \Delta y \Delta z$



All results are assembled to yield:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} \doteq \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \underbrace{\Delta x \Delta y \Delta z}_{\Delta v} = Q \quad (\text{by Gauss' Law})$$

where Q is the charge enclosed within volume Δv

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



- Divergence 散度

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \underbrace{\Delta x \Delta y \Delta z}_{\Delta v} = Q \quad (\text{by Gauss' Law})$$

where Q is the charge enclosed within volume Δv

The divergence of the vector flux density $\mathbf{A}$ is the outflow of flux from a small closed surface per unit volume as the volume shrinks to zero.

当闭合曲面所包围的体积趋近于0时，通量密度 $\mathbf{A}$ 的散度则表示从单位体积的表面穿出的通量。

Mathematically, this is:

$$\text{Divergence of } \mathbf{A} = \text{div } \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

7: Divergence, Maxwell's First Equation

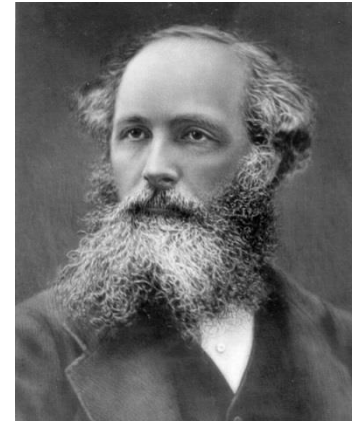
散度、麦克斯韦第一方程



- Maxwell's first equation 麦克斯韦第一方程

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \underbrace{\Delta x \Delta y \Delta z}_{\Delta v} = Q \quad (\text{by Gauss' Law})$$

$$\text{Divergence of } \mathbf{A} = \text{div } \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$



James Clerk Maxwell
(1831–1879)

and when the vector field is the electric flux density:

$$\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{\Delta v} = \lim_{\Delta v \rightarrow 0} \frac{Q}{\Delta v} = \rho_v = \underline{\text{div } \mathbf{D}}$$

$$\text{div } \mathbf{D} = \rho_v$$

Maxwell's first equation

Note: Essentially, Maxwell's first equation is the differential form of Gauss's Law.

本质上，麦克斯韦第一方程是高斯定律的一种微分表达形式。

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



- Divergence Expressions in the Three Coordinate Systems 散度如何计算？在三种不同坐标系中的表达

$$\text{Divergence of } \mathbf{D} = \text{div } \mathbf{D} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{\Delta v}$$

$$\text{div } \mathbf{D} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \quad (\text{rectangular})$$

$$\text{div } \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \quad (\text{cylindrical})$$

$$\text{div } \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_\theta) + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi} \quad (\text{spherical})$$

Note: 1) 只有通量密度才能定义散度，而通量密度一定是矢量；
2) 矢量的散度是标量。

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



- The Del Operator 矢量Del算子

The del operator is a *vector differential operator*, and is defined as:

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

Note that:

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \cdot (D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z) \\ &= \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \text{div } \mathbf{D} \end{aligned}$$

$$\nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \text{div}(\mathbf{D})$$

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



- Divergence Theorem 散度定理

We now have Maxwell's first equation (or the point form of Gauss' Law) which states:

$$\operatorname{div} \mathbf{D} = \nabla \cdot \mathbf{D} = \rho_v$$

and Gauss's Law in large-scale form reads:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

leading to the Divergence Theorem:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程

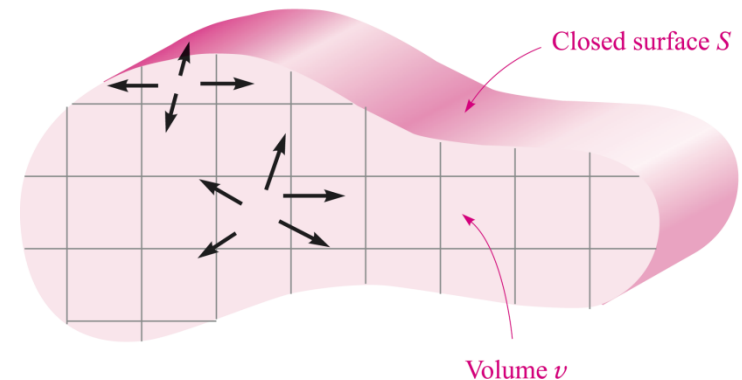


- Divergence Theorem 散度定理

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed by the closed surface.

任意矢量场的法向分量在闭合面上的面积分等于它的散度在闭合面所包围的体积内的体积分。

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$



Note: Divergence Theorem is applicable to any vector field.

散度定理适用于任意矢量场。

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程

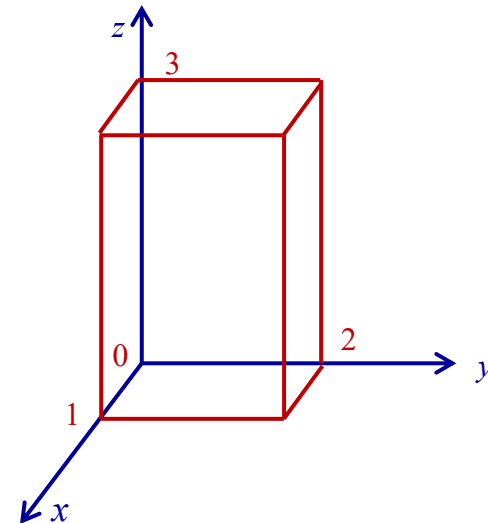


- Example:

Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\mathbf{a}_x + x^2\mathbf{a}_y$ C/m² and the rectangular parallelepiped formed by the planes $x = 0$ and 1 , $y = 0$ and 2 , and $z = 0$ and 3 .

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

Solution. Evaluating the surface integral first, we note that $\mathbf{D}$ is parallel to the surfaces at $z = 0$ and $z = 3$, so $\mathbf{D} \cdot d\mathbf{S} = 0$ there. For the remaining four surfaces we have



$$\begin{aligned} \oint_S \mathbf{D} \cdot d\mathbf{S} &= \int_0^3 \int_0^2 (\mathbf{D})_{x=0} \cdot (-dy dz \mathbf{a}_x) + \int_0^3 \int_0^2 (\mathbf{D})_{x=1} \cdot (dy dz \mathbf{a}_x) \\ &\quad + \int_0^3 \int_0^1 (\mathbf{D})_{y=0} \cdot (-dx dz \mathbf{a}_y) + \int_0^3 \int_0^1 (\mathbf{D})_{y=2} \cdot (dx dz \mathbf{a}_y) \end{aligned}$$

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



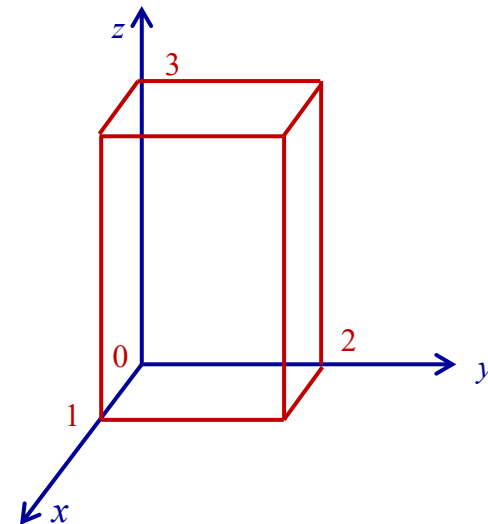
- Example:

Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\mathbf{a}_x + x^2\mathbf{a}_y$ C/m² and the rectangular parallelepiped formed by the planes $x = 0$ and 1, $y = 0$ and 2, and $z = 0$ and 3.

$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{S} &= \int_0^3 \int_0^2 (\mathbf{D})_{x=0} \cdot (-dy dz \mathbf{a}_x) + \int_0^3 \int_0^2 (\mathbf{D})_{x=1} \cdot (dy dz \mathbf{a}_x) \\ &\quad + \int_0^3 \int_0^1 (\mathbf{D})_{y=0} \cdot (-dx dz \mathbf{a}_y) + \int_0^3 \int_0^1 (\mathbf{D})_{y=2} \cdot (dx dz \mathbf{a}_y) \\ &= - \int_0^3 \int_0^2 (D_x)_{x=0} dy dz + \int_0^3 \int_0^2 (D_x)_{x=1} dy dz \\ &\quad - \int_0^3 \int_0^1 (D_y)_{y=0} dx dz + \int_0^3 \int_0^1 (D_y)_{y=2} dx dz\end{aligned}$$

However, $(D_x)_{x=0} = 0$, and $(D_y)_{y=0} = (D_y)_{y=2}$, which leaves only

$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{S} &= \int_0^3 \int_0^2 (D_x)_{x=1} dy dz = \int_0^3 \int_0^2 2y dy dz \\ &= \int_0^3 4 dz = 12\end{aligned}$$



$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



- Example:

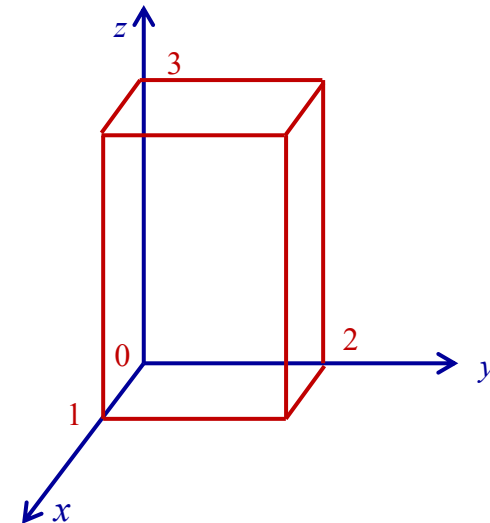
Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\mathbf{a}_x + x^2\mathbf{a}_y$ C/m² and the rectangular parallelepiped formed by the planes $x = 0$ and 1 , $y = 0$ and 2 , and $z = 0$ and 3 .

Since

$$\nabla \cdot \mathbf{D} = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(x^2) = 2y$$

the volume integral becomes

$$\begin{aligned} \int_{\text{vol}} \nabla \cdot \mathbf{D} dv &= \int_0^3 \int_0^2 \int_0^1 2y dx dy dz = \int_0^3 \int_0^2 2y dy dz \\ &= \int_0^3 4 dz = 12 \end{aligned}$$



and the check is accomplished. Remembering Gauss's law, we see that we have also determined that a total charge of 12 C lies within this parallelepiped.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

7: Divergence, Maxwell's First Equation

散度、麦克斯韦第一方程



D3.9. Given the field $\mathbf{D} = 6\rho \sin \frac{1}{2}\phi \mathbf{a}_\rho + 1.5\rho \cos \frac{1}{2}\phi \mathbf{a}_\phi$ C/m², evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\phi = 0$, $\phi = \pi$, $z = 0$, and $z = 5$.

HW-3



3. The cylindrical surface $\rho = 8$ cm contains the surface charge density, $\rho_s = 5e^{-20|z|}$ nC/m². a) What is the total amount of charge present? b) How much flux leaves the surface $\rho = 8$ cm, $1 \text{ cm} < z < 5 \text{ cm}$, $30^\circ < \varphi < 90^\circ$?
4. Let $\mathbf{D} = 4xy\mathbf{a}_x + 2(x^2 + z^2)\mathbf{a}_y + 4yz\mathbf{a}_z$ C/m² and evaluate surface integrals to find the total charge enclosed in the rectangular parallelepiped $0 < x < 2$, $0 < y < 3$, $0 < z < 5$ m.
5. An infinitely long cylindrical dielectric of radius b contains charge within its volume of density $\rho_v = a\rho^2$, where a is a constant. Find the electric field strength, $\mathbf{E}$, both inside and outside the cylinder.